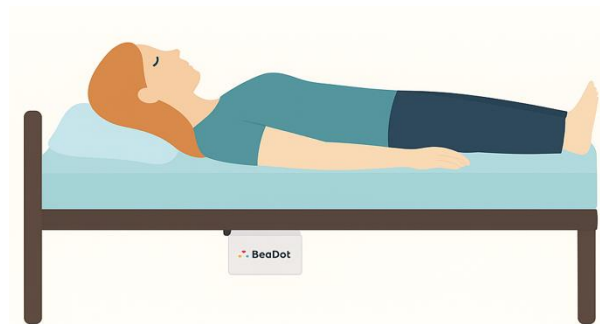




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Multi-granularity Supervised Contrastive Learning with Online Adaptation for **Contactless** In-bed Posture Classification

Authors: **Yingjian Song**, Haotian Xiang, Zixuan Zeng, Jiayu Chen, Yida Zhang, Zaid Farooq Pitafi, He Yang, Qin Lu, Xiang Zhang, Bradley G Phillips, **Fei Dou**, **WenZhan Song**

Presenter: **Fei Dou**, Assistant Professor, School of Computing, University of Georgia

BedDot: Engagement-free Monitoring of Sleep Activities and Vital Signs

Features

• Vital signs

- Heart rate (+/- 3 bpm)
- Respiration rate (+/- 3 bpm)
- Blood pressure (+/- 5 mmHg)
- Inter-beat-interval (+/-20 ms)

• Sleep activities

- Movements
- Posture changes

• Important events

- Bed occupancy



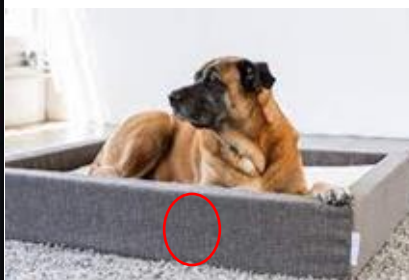
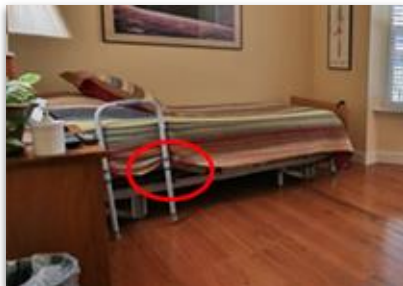
National Heart, Lung, and Blood Institute



GRA Venture Fund



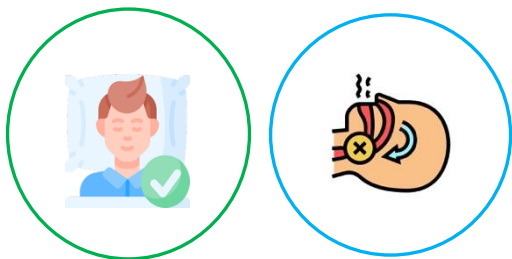
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Intelligent Dots

Introduction

Why Sleep Posture Recognition



Improved Sleep Quality Sleep Disorders Health

- Existing modalities: privacy, setup, or radiation issues.



	Modality	Setup	Privacy	Intrusive	Interference
	BCG	Complicated	High	Low	Low
	RF	Complicated	High	Low	High
	Camera	Complicated	Low	Low	Low
	Wearable	Easy	High	High	Low
	Seismic	Easy	High	Low	Low

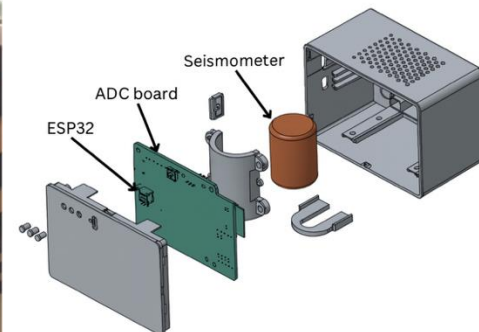
Insight: Bodyseismography (BSG) is easy to install; privacy-friendly; non-intrusive and contactless; passive (no active radiations).

Hardware & Software Solution

- **Hardware-BedDot:** ESP32 + 100 Hz 24bit ADC + 4.5Hz geophone sensor + under bed magnetic mount
- **Streaming:** MQTT → InfluxDB → Grafana
- **Features:**
 - Vital signs:
 - heart rate, IBI, HRV
 - respiration rate
 - blood pressure
 - Bed occupancy¹
 - Posture changes (this paper)



Installation of BedDot under the bed frame

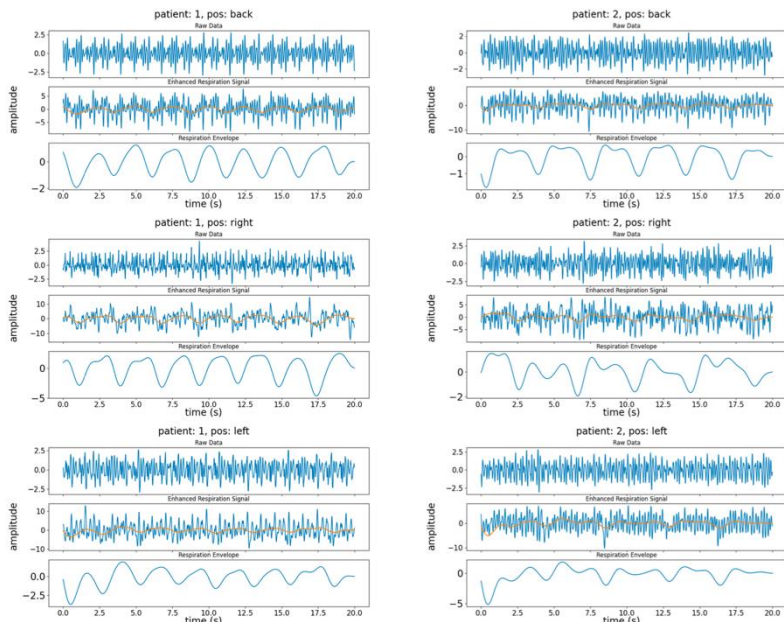


BedDot device



Challenges & Motivation I: Subject & Posture Distribution Shift

■ Variability Across Postures and Individuals:



Respiratory and heartbeat patterns differ across postures (rows) and individuals (columns).

■ Challenges

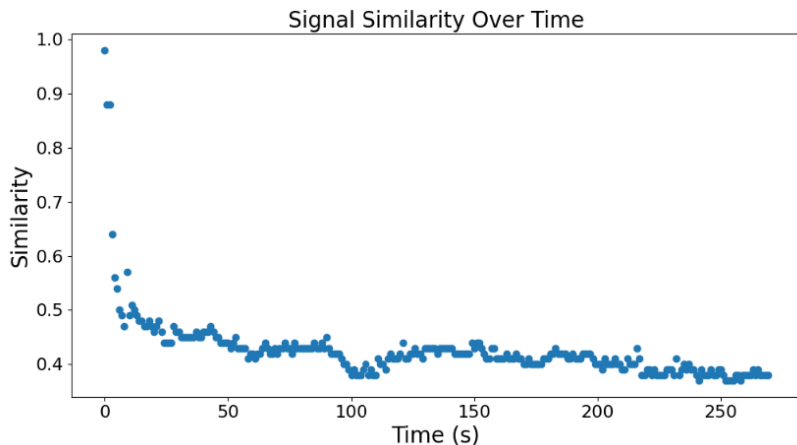
- **posture-dependent** signal **variations**:
 - Respiration and cardiac waveforms change
- **subject-dependent** **variability**:
 - *same posture*: amplitude & morphology differ
- → **cross-domain distribution shifts** unseen
 - static models **fail to generalize** to unseen

- **Multi-Granularity Learning**: Learn **posture-invariant yet subject-adaptive** representations; Align local and global spatial features across domains.

Key takeaway: builds domain-robust embeddings to generalize across subjects and postures.

Challenges & Motivation II: Temporal Distribution Shift

■ Changes in Signal Distribution Over Time:



Similarity between the initial second and subsequent signals under the same posture declines over time.

■ Challenges:

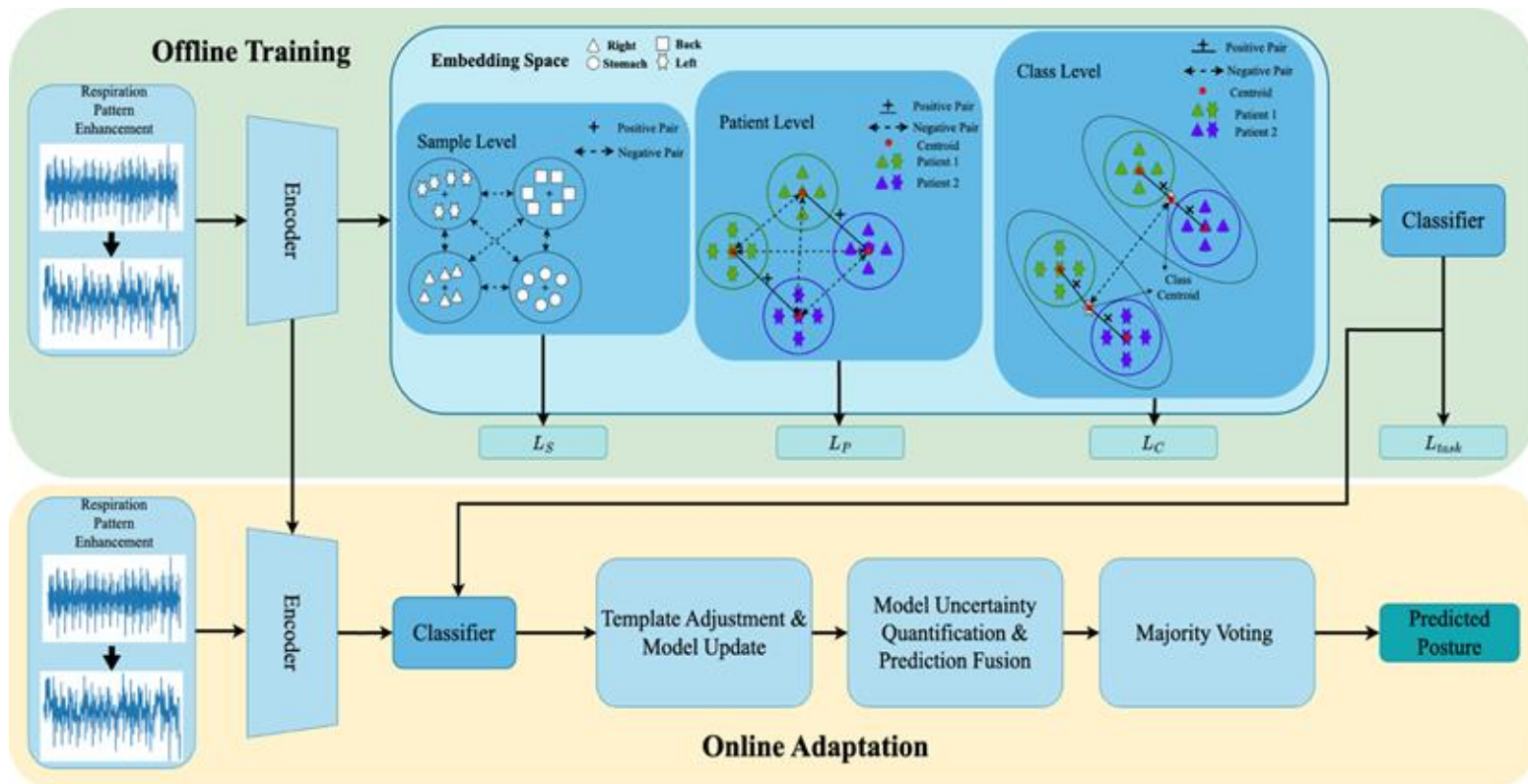
- **same posture, signal similarity decreases**
- indicates a **temporal drift** caused by:
 - Sensor noise and temperature variation
 - Micro body movements or bed mechanics
 - Physiological rhythm changes (breathing, heartbeat)
- drift → **distribution mismatch**

■ Test-Time Adaptation (TTA) to continuously **recalibrate** model to evolving signal patterns.

Key takeaway: static models degrade over time — dynamic adaptation maintains robustness.

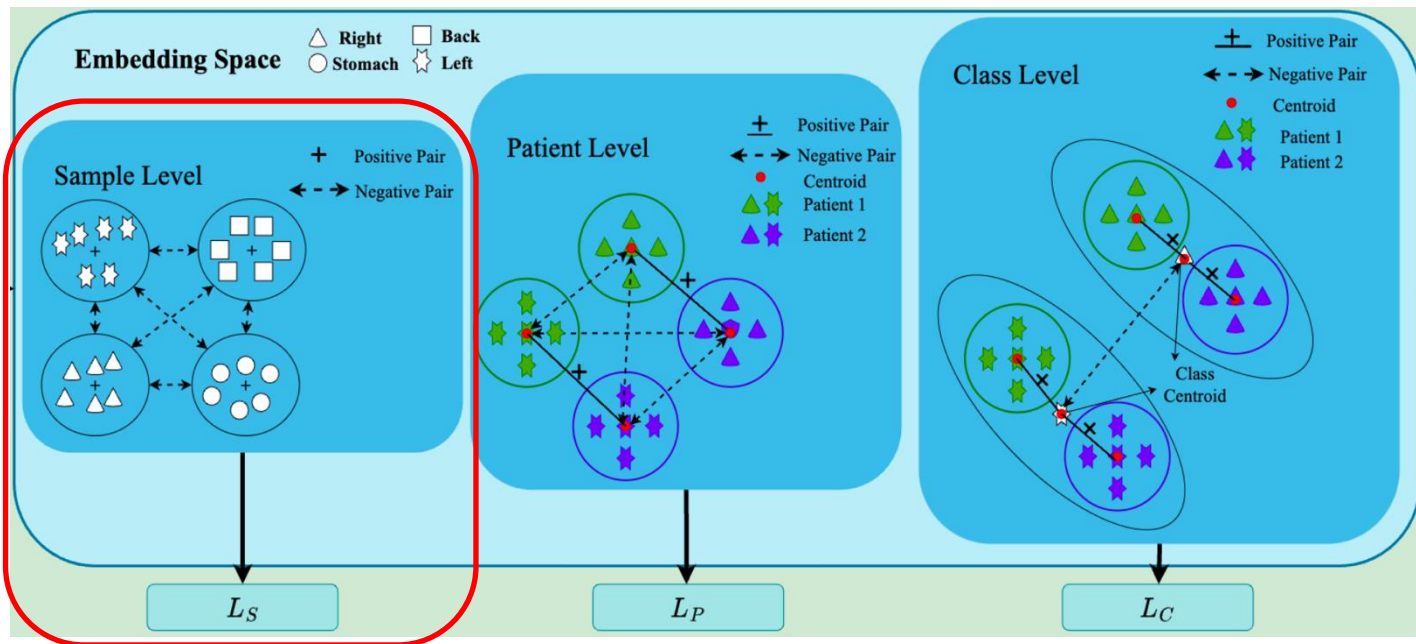
MGSCLoa for In-bed Posture Prediction

- Multi-Granularity Supervised Contrastive Learning (MGSCCL) with Online Adaptation



MGSCLoa: Offline Training-MGSCL

- Multi-Granularity Supervised Contrastive Learning (SSCL, PSCL, CSCL) + task loss

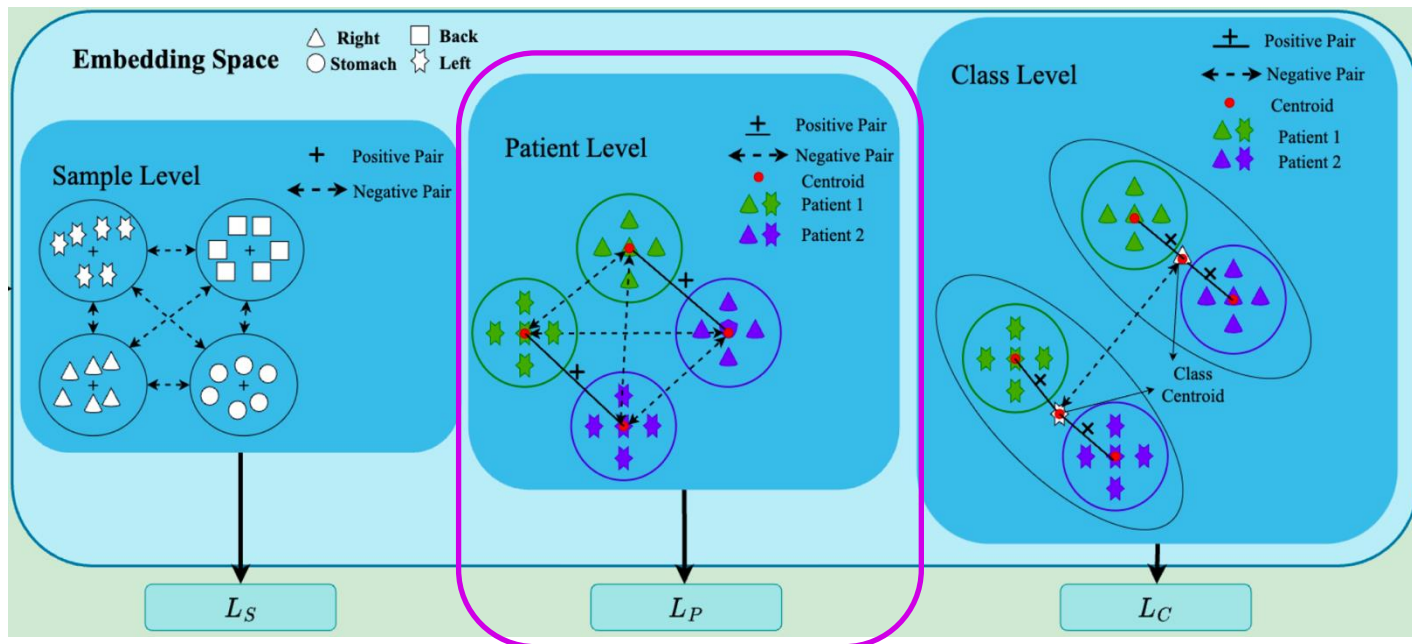


- Sample-level (**SSCL**): maximize similarity within class; minimize across classes

$$L_S(\theta) = -\frac{1}{N} \sum_{n=1}^N \frac{1}{|I_{y_n}| - 1} \sum_{i \in I_{y_n} \setminus \{n\}} \log \frac{\exp(\mathbf{z}_n^\top \mathbf{z}_i / \tau)}{\sum_{i' \in \bar{I}_{y_n}} \exp(\mathbf{z}_n^\top \mathbf{z}_{i'} / \tau)}.$$

MGSCLoa: Offline Training-MGSCL

- Multi-Granularity Supervised Contrastive Learning (SSCL, PSCL, CSCL) + task loss



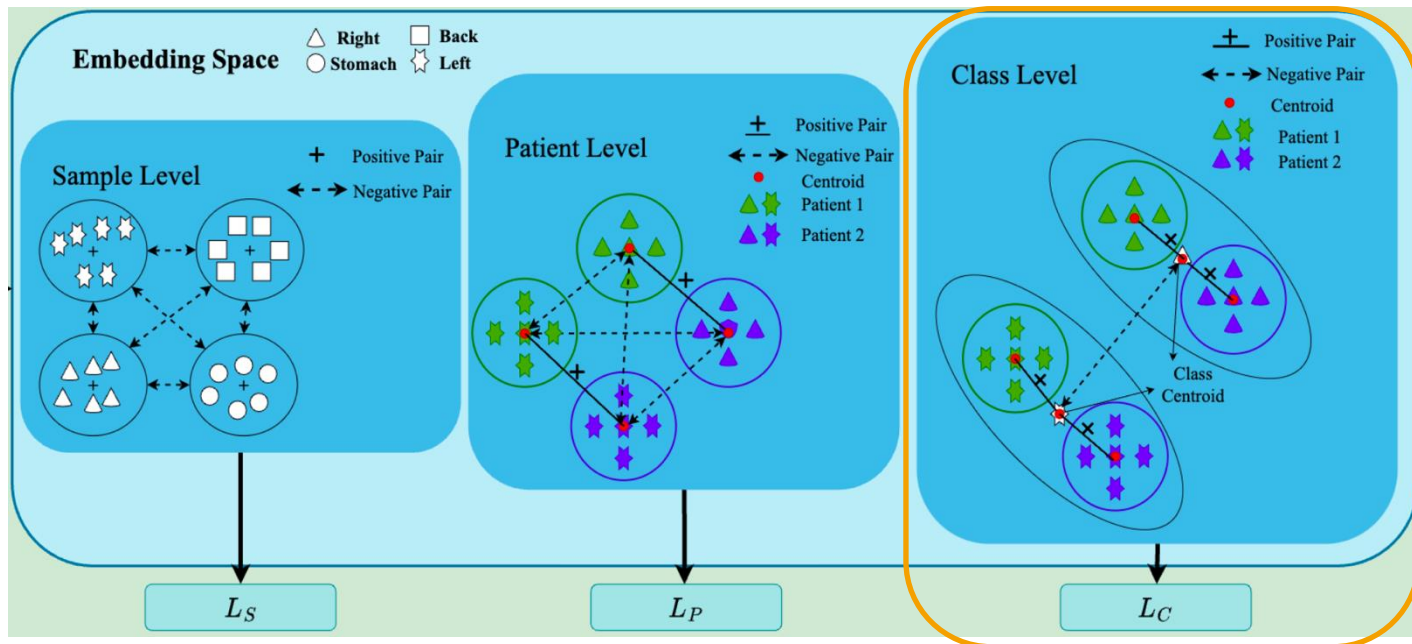
- Patient-level (**PSCL**): pull same-class centroids across patients, push different classes

$$\text{PSCL Centroids: } \bar{\mathbf{z}}_{p,c} = \frac{1}{|I_c^p|} \sum_{n \in I_c^p} \mathbf{z}_n$$

$$L_P(\theta) = -\frac{1}{PC} \sum_{p=1}^P \sum_{c=1}^C \frac{1}{|\mathcal{P}| - 1} \sum_{p' \in \mathcal{P} \setminus \{p\}} \log \frac{\exp(\bar{\mathbf{z}}_{p,c}^\top \bar{\mathbf{z}}_{p',c} / \tau)}{\sum_{p' \in \mathcal{P}} \sum_{c' \in \mathcal{C} \setminus \{c\}} \exp(\bar{\mathbf{z}}_{p,c}^\top \bar{\mathbf{z}}_{p',c'})}$$

MGSCLoa: Offline Training-MGSCL

- Multi-Granularity Supervised Contrastive Learning (SSCL, PSCL, CSCL) + task loss



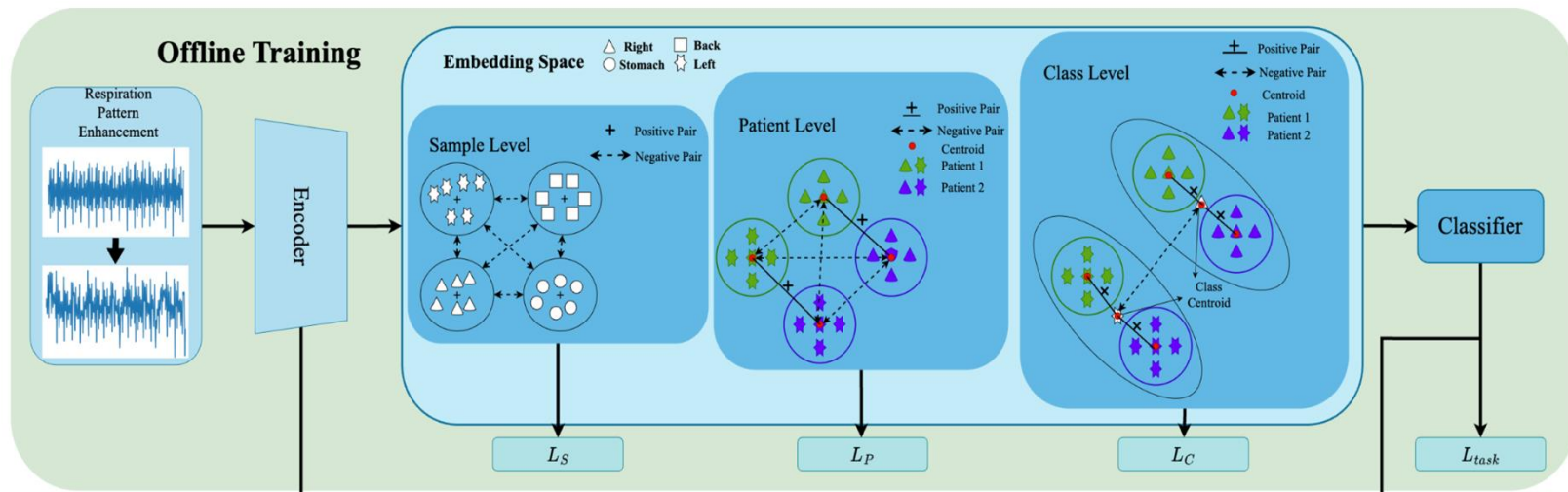
- Class-level (**CSCL**): align patient centroids to class centroid; repel other classes

Class centroids: $\bar{\mathbf{z}}_c = \frac{1}{|I_c|} \sum_{i \in I_c} \mathbf{z}_i$

$$L_C(\theta) = -\frac{1}{C} \sum_{c \in C} \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \log \frac{\exp(\bar{\mathbf{z}}_c^\top \bar{\mathbf{z}}_{p,c}/\tau)}{\sum_{c' \in C \setminus \{c\}} \exp(\bar{\mathbf{z}}_c^\top \bar{\mathbf{z}}_{c'})}$$

MGSCLoa: Offline Training-MGSCL

- Multi-Granularity Supervised Contrastive Learning (SSCL, PSCL, CSCL) + task loss



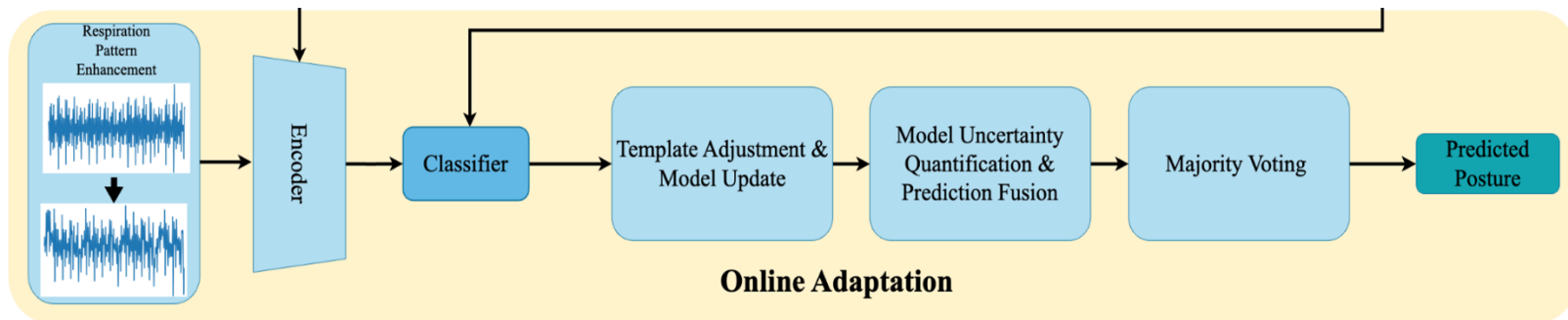
Overall:

$$L(\theta, \phi) = \lambda_1 L_C + \lambda_2 L_P + \lambda_3 L_S + \lambda_4 L_{task}$$

Recommended weights: (0.1, 0.1, 1, 1); task loss is cross-entropy

MGSCLoa: Online Adaptation (oa)

- Online Adaptation (oa): SPD, TAMU, UQF, MJV

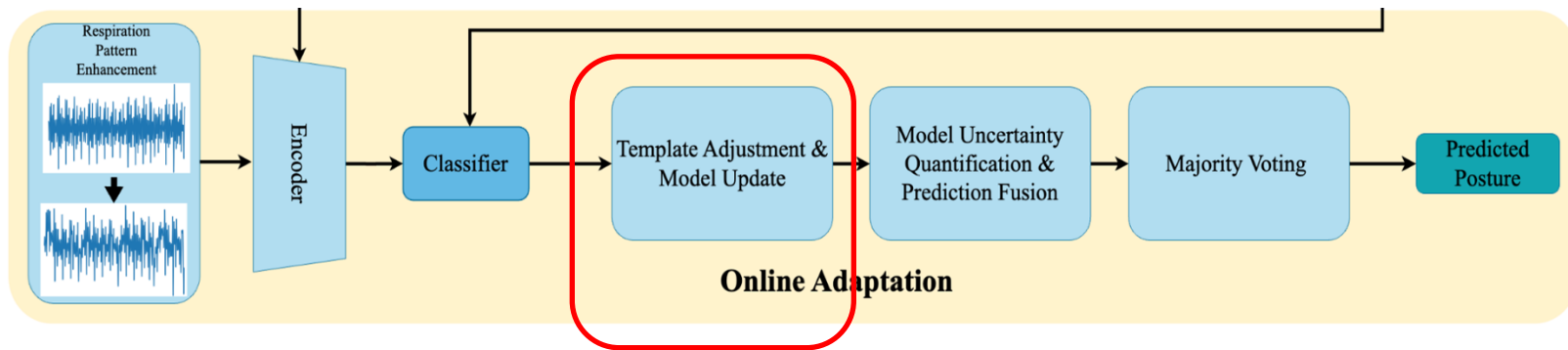


- Stationary Period Detection (SPD): detect movement via kurtosis; threshold $\tau_{\kappa}=7$

$$\kappa_t = \sum_{d=1}^D \frac{(x_t(d) - \bar{x}_t(d))^4}{\sigma_t^4}.$$

MGSCLoa: Online Adaptation (oa)

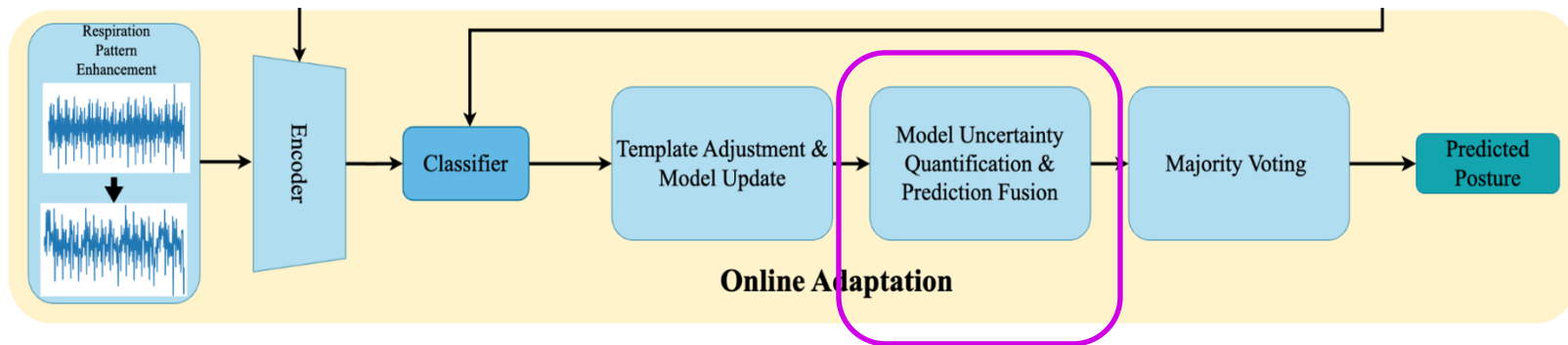
- Online Adaptation (oa): SPD, TAMU, UQF, MJV



- Template Adjustment and Model Update (**TAMU**): entropy sharpness minimization + support set expansion; update extractor + classifier:
 - Sharpness-Aware Entropy Minimization:
 - minimizes prediction entropy and sharpness; filtering out uncertain samples
 - Template Adjustment:
 - gradually updates posture templates with confident ones

MGSCLoa: Online Adaptation (oa)

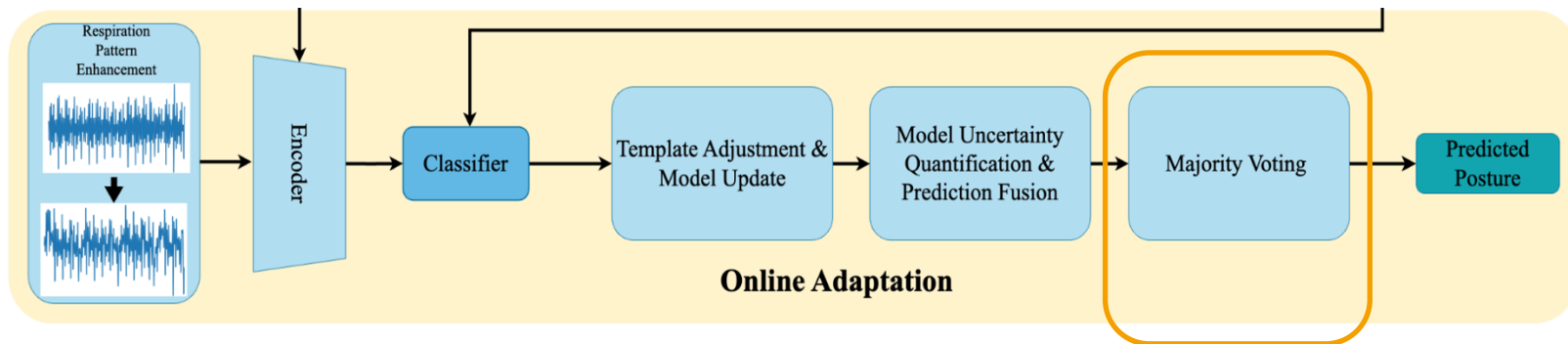
- Online Adaptation (oa): SPD, TAMU, UQF, MJV



- Model Uncertainty Quantification and Prediction Fusion (**UQF**): dropout ensemble; variance-based weighting; fuse with base model
 - enabling Dropout layers to produce **multiple** different **predictions**
 - calculate **variance** of the prediction for each model during the stationary period
 - assign weights to each model based on the variance
 - ensemble** the final prediction

MGSCLoa: Online Adaptation (oa)

- Online Adaptation (oa): SPD, TAMU, UQF, MJV



- Majority Voting (**MJV**): majority vote within stationary segment

Datasets & Protocol

- Phase 1 (clinical):
 - Hospital Bed
 - 100 subjects;
 - back/right/left/stomach
 - 20 s windows;
 - 5-fold CV.
- Phase 2 (home):
 - 11 participants;
 - 13 nights;
 - 3 min per common posture for few-shot.

Cross-subjects: no same patients between the training and testing

Table 13. Bed Types for Phase Two Data Collection

Environment	Bed Size	Bed Type	Frame Material
Home1	Queen	Foam	Metal
Home2	Queen	Foam	Metal
Home3	King	Foam	Metal
Home4	King	Spring	Metal
Home5	King	Foam	Wood
Home6	Queen	Foam	Wood
Home7	Queen	Foam	Wood
Home8	Queen	Foam	Wood
Home9	Twin XL	Thick Foam	Metal
Home10	Queen	Foam	Metal
Home11	Queen	Spring	Metal

Phase 1: Overall Performance

Overall Performance of MGSCLoa (Acc stands for accuracy, both Acc and F1 are in %). Underline is the second best result and bold is the best result.

METHOD	Metric	Fold1	Fold2	Fold3	Fold4	Fold5	Ave
MGSCl ($\mathcal{D}_{tr}$)	Acc	68.48	73.59	69.59	73.52	72.78	71.59 $\pm$ 2.14
	F1	67.97	73.45	69.61	73.47	72.66	71.43 $\pm$ 2.24
MGSCl ($\mathcal{D}_{test}$)	Acc	67.64	72.74	68.55	71.78	73.11	70.76 $\pm$ 2.24
	F1	66.92	72.51	68.55	71.87	73.10	70.59 $\pm$ 2.42
MGSCl + OA ($\mathcal{D}_{test}$)	Acc	<u>81.47</u>	<u>81.89</u>	<u>78.28</u>	<u>81.39</u>	<u>85.53</u>	<u>81.71 $\pm$ 2.30</u>
	F1	<u>81.08</u>	<u>81.82</u>	<u>78.18</u>	<u>81.32</u>	<u>85.25</u>	<u>81.53 $\pm$ 2.25</u>
MGSCl + OA + FS ($\mathcal{D}_{test}$)	Acc	94.89	93.28	91.99	87.30	90.89	91.67 $\pm$ 2.56
	F1	94.88	93.23	91.92	87.25	90.36	91.53 $\pm$ 2.61

* Few-shot uses first 40s per posture/patient (20 samples)

Overall Performance of MGSCLoa on each posture

METHOD	Metric	Back	Right	Left	Stomach
MGSCl ($\mathcal{D}_{tr}$)	Acc	78.54 $\pm$ 8.85	66.96 $\pm$ 8.96	63.49 $\pm$ 10.36	77.14 $\pm$ 5.19
	F1	79.54 $\pm$ 5.30	63.50 $\pm$ 5.61	65.39 $\pm$ 5.93	77.30 $\pm$ 3.69
MGSCl ($\mathcal{D}_{test}$)	Acc	77.87 $\pm$ 7.60	66.89 $\pm$ 7.33	62.19 $\pm$ 10.36	75.84 $\pm$ 5.46
	F1	79.16 $\pm$ 4.72	62.30 $\pm$ 4.21	64.26 $\pm$ 6.33	76.64 $\pm$ 3.54
MGSCl + OA ($\mathcal{D}_{test}$)	Acc	<u>85.80 $\pm$ 7.74</u>	<u>72.06 $\pm$ 4.52</u>	<u>77.08 $\pm$ 6.08</u>	<u>89.13 $\pm$ 5.85</u>
	F1	<u>85.44 $\pm$ 6.31</u>	<u>74.10 $\pm$ 2.72</u>	<u>77.77 $\pm$ 4.10</u>	<u>86.26 $\pm$ 4.95</u>
MGSCl + OA + FS ($\mathcal{D}_{test}$)	Acc	98.21$\pm$1.42	80.55$\pm$8.07	90.21$\pm$2.56	94.59$\pm$5.74
	F1	95.07$\pm$1.94	87.24$\pm$4.45	90.92$\pm$3.82	89.97$\pm$4.60

+10.9 % $\uparrow$ (OA vs no OA)

+20.9 % $\uparrow$ (FS vs baseline)

■ Interpretation

- MGSCl improves **cross-subject and cross-posture** robustness.
- OA **mitigates temporal distribution drift** via test-time adaptation.
- FS personalizes model to new patients **with minimal labeled data** (40 s per posture).

Cross-subjects: no same patients
between the training and testing

Key takeaway: MGSCLoa achieves 91.7 % accuracy and 91.5 % F1, surpassing baseline by > 20 %. The combination of *multi-granularity contrastive learning* and *online adaptation* **ensures both cross-subject and temporal robustness.**

Phase 1 : vs Other Modalities

■ Baseline Systems & Modalities

- BCG, RF/mmWave, Wi-Fi, Camera, Wearables
- **Require body contact / complex setup / privacy concerns**

■ Our Results

- MGSCCL → 71.6 % Acc / 71.4 % F1
- MGSCLoa (+ OA) → 81.7 % Acc / 81.5 % F1
- Outperforms best radar-based method (+9.6 %)

■ Interpretation

- Seismic sensor offers a **non-intrusive, privacy-preserving** solution.
- MGSCCL **ensures cross-patient robustness**.
- OA ensures **temporal adaptation** in real deployment.

Comparison between existing sleep posture classification system and MGSCCL

	METHOD	Metric	Fold1	Fold2	Fold3	Fold4	Fold5	Ave
BCG - NN	[51]	Acc	31.98	30.77	31.95	30.98	29.91	31.12 ± 0.87
		F1	27.66	24.94	25.37	30.92	25.21	26.82 ± 2.54
Radar - DT	[46]	Acc	37.54	36.47	32.10	36.56	33.86	35.30 ± 2.25
		F1	36.85	32.30	29.16	33.62	30.67	32.52 ± 2.95
WiFi - RF	[24]	Acc	41.04	38.17	29.17	40.45	36.12	36.99 ± 4.79
		F1	36.20	31.60	24.64	35.67	29.54	31.53 ± 4.75
BCG-FFN	[8]	Acc	37.02	34.49	30.33	34.91	32.28	33.81 ± 2.57
		F1	34.18	29.92	28.58	30.77	30.09	30.71 ± 2.10
mmWave NN	[54]	Acc	63.61	59.71	57.27	66.19	63.03	61.96 ± 3.49
		F1	63.96	57.38	56.03	66.15	60.66	60.84 ± 4.27
Radar Wavelet	[40]	Acc	36.61	30.38	37.94	36.80	37.68	35.88 ± 3.13
		F1	32.40	25.99	36.66	30.57	33.31	31.78 ± 3.92
Ours	MGSCCL	Acc	<u>68.48</u>	<u>73.59</u>	<u>69.59</u>	<u>73.52</u>	<u>72.78</u>	<u>71.59 ± 2.39</u>
		F1	<u>67.97</u>	<u>73.45</u>	<u>69.61</u>	<u>73.47</u>	<u>72.66</u>	<u>71.43 ± 2.50</u>
Ours	MGSCLoa	Acc	81.47	81.89	78.28	81.39	85.53	81.71 ± 2.58
		F1	81.08	81.82	78.18	81.32	85.25	81.53 ± 2.52

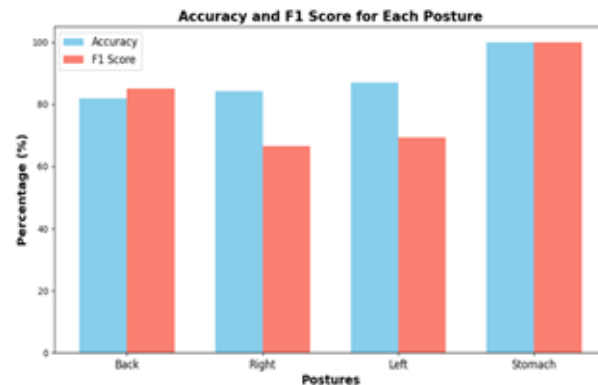
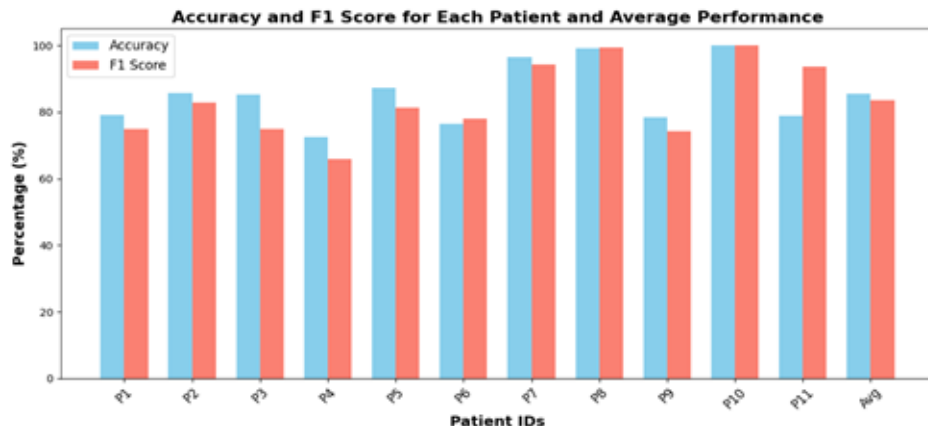
Key takeaway: MGSCLoa achieves > 80 % accuracy—**outperforming other** existing **sensor modalities** while remaining easy to deploy and privacy-friendly.

Phase 1 : Cross-methods; Ablations & Sensitivity

- **VS. contrastive baselines:** MGSCl > SCL by +2.91 acc / +2.88 F1 (avg).
- **VS. TTA baselines:** OA +10.95 acc / +10.94 F1 over MGSCl; TAMU alone +1.36 / +1.38; Tent/OFTTA degrade.
- **Contrastive pieces:** +S:+1.02/+1.27; +P:+2.12/+2.14; +C:+1.48/+1.65; all best (71.59/71.43)
- **OA pieces:** MJV adds ~6–8% across settings
- **Extractor:** STCN 71.59 vs Transformer 31.64 acc
- **Bandpass:** 0.1–20 Hz best avg
- **Weights:** best [s, p, c] = [0.1, 0.1, 0.3]; stable overall

Key takeaway: **Each module contributes complementary gains**, and the framework remains stable across configurations, confirming the robustness and reliability of MGSCLoa.

Phase 2 Real-Sleep Scenario - Cross-Environment Generalization



Experiment Setup

- 11 participants $\times$ 13 nights $\approx$ 80 h real sleep data
- 3 min** labeled samples per posture (few-shot)
- Training: Phase 1 + leave-one-subject-out Phase 2

Results

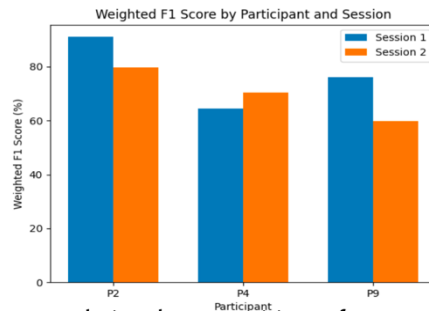
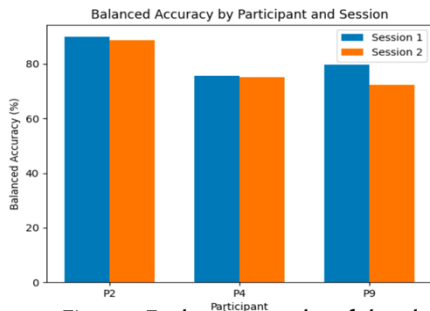
- Acc 85.37 %, F1 83.59 %**
- All postures > 80% accuracy
- $\sim$ 6% drop from controlled setup $\rightarrow$ robust generalization

Insights

- Minimal supervision** achieves strong personalization
- Robust to home environment noise and bed variability**
- Confirms **real-world deployability** of MGSCLoa

Key takeaway: MGSCLoa maintains >85% accuracy in real-sleep conditions with minimal calibration, demonstrating robust **cross-domain adaptability and real-world readiness**.

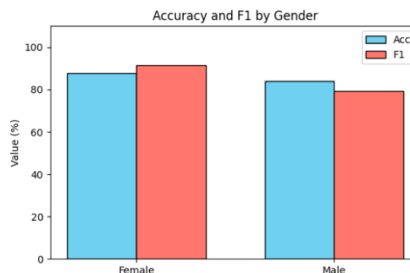
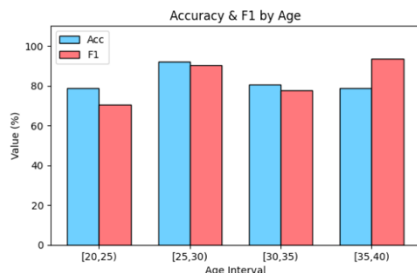
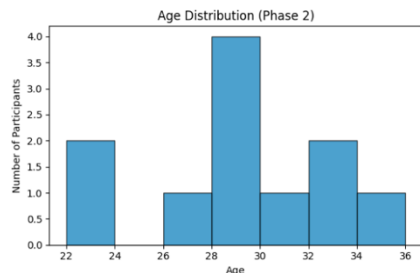
Phase 2: Robustness to Sensor & Demographic Variability



Investigation of Sensor Position

■ Sensor Position Robustness

- Evaluated across nights and re-installations
- Accuracy stable (80–85%), ≤5% variation
- Demonstrates **spatial robustness to installation drift**



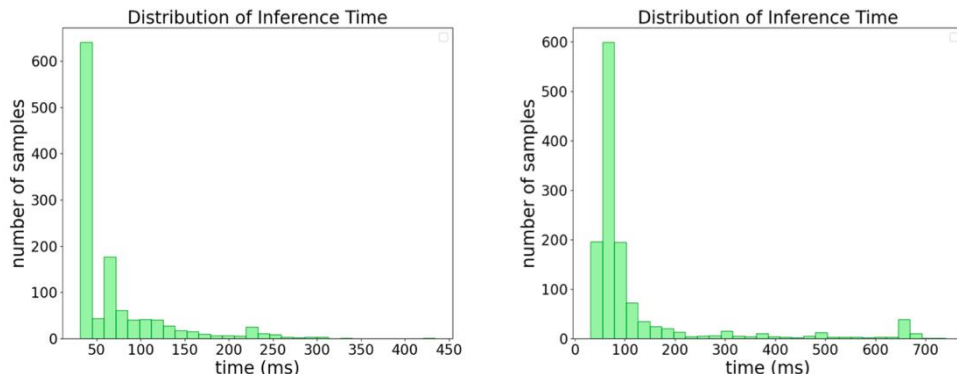
Phase 2 Age distribution (left) and mean accuracy and F1 by different Age groups (middle), and mean Accuracy and F1 by different gender (right).

■ Demographic Robustness

- Tested **across gender and age** subgroups (22–35 yrs)
- **No significant performance gap** (<2%)
- Confirms fairness and generalization across users

Key takeaway: MGSCLoa remains **stable across sensor positions and user demographics**, proving its robustness and fairness for real-world deployment.

Computational Efficiency and Real-Time Capability



Computational Time distribution for Phase 1 data (left), and Computational Time distribution for Phase 2 data (right).

Key takeaway: MGSCLoa achieves **real-time inference (<0.5 s)** with low computational overhead, enabling practical, continuous, and privacy-preserving sleep posture monitoring.

■ Setup

- 20 s window / 1 s stride
- AMD Ryzen 7985WX + RTX 6000
- Online streaming inference

■ Results

- ICU: 397 ± 354 ms
- Phase 1: 69.9 ± 54.9 ms
- Phase 2: 122.8 ± 144.6 ms
- **< 0.5 s per prediction → real-time capable**

■ Implications

- **Efficient** for continuous sleep monitoring
- **Scalable** to edge deployment
- No perceptible delay in real applications

Discussions & Future Directions

- Dataset & Environment
 - Data sets (360+ subjects): ~100 ICU patients, ~100 sleep lab patients, ~160 clinical trial subjects
 - Future: broader bed types & environmental diversity
- Generalization
 - MGSCl adaptable to other time-series tasks
 - OA modules (UQF/TAMU/SPD) generalizable under stationary conditions
- Limitations & Future Work
 - Smartphone accelerometer labeling → multi-sensor validation
 - Sensor placement sensitivity → tri-axis & spatial modeling
 - Hyperparameter robustness ($\sigma < 2\%$) → easy real-world deployment

Key takeaway: MGSCLoa provides a **generalizable, adaptive, and efficient foundation** for time-series learning — ready to expand beyond posture classification into broader physiological monitoring.

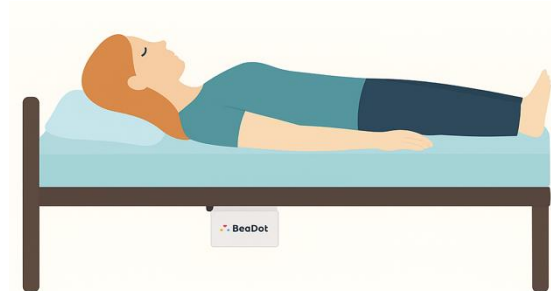
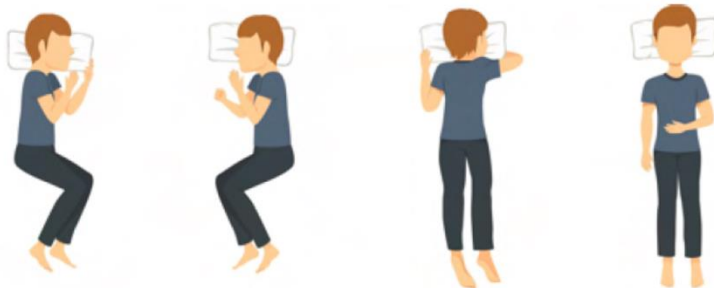
Takeaways

- Multi-granularity supervised contrastive learning aligns cross-patient clusters.
- Online adaptation leveraging stationary periods yields large gains.
- **BSG** is privacy-friendly and deployable in clinic & home, in **contactless** manner.

Thank you for listening!



Questions or Comments



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[UbiComp'26] Jiayu Chen, Yingjian Song, Yida Zhang, Zixuan Zeng, Xiang Zhang, Zaid Pitafi, Zaipeng Xie, Deepak Kumar Das, Nishan Dong, Junjie Lu, Xiao Yin, WenZhan Song, "SelfDenoiser: Self-supervised Seismic Signal Denoiser for Continuous and Contactless Cardiac Monitoring", ACM UbiComp/IMWUT 2026

[UbiComp'25] Yingjian Song, Haotian Xiang, Zixuan Zeng, Jiayu Chen, Yida Zhang, Zaid Farooq Pitafi, He Yang, Qin Lu, Xiang Zhang, Brad Phillips, Fei Dou, WenZhan Song, "Multi-Granularity Supervised Contrastive Learning with Online Adaptation for Contactless In-Bed Posture Classification", ACM UbiComp/IMWUT 2025

[UbiComp'24] Yingjian Song, Zaid Farooq Pitafi, Fei Dou, Jin Sun, Xiang Zhang, Bradley G. Phillips, WenZhan Song, "Self-Supervised Representation Learning and Temporal-Spectral Feature Fusion for Bed Occupancy Detection", ACM UbiComp/IMWUT 2024

[PerCom'20] Jose Clemente; Maria Valero; Fangyu Li; Chengliang Wang; WenZhan Song; Helena: Real-time Contact-free Monitoring of Sleep Activities and Events around the Bed, 18th IEEE Conference on Pervasive Computing and Communications (PerCom 2020), 2020.

[Mark Weiser Best Paper Award]